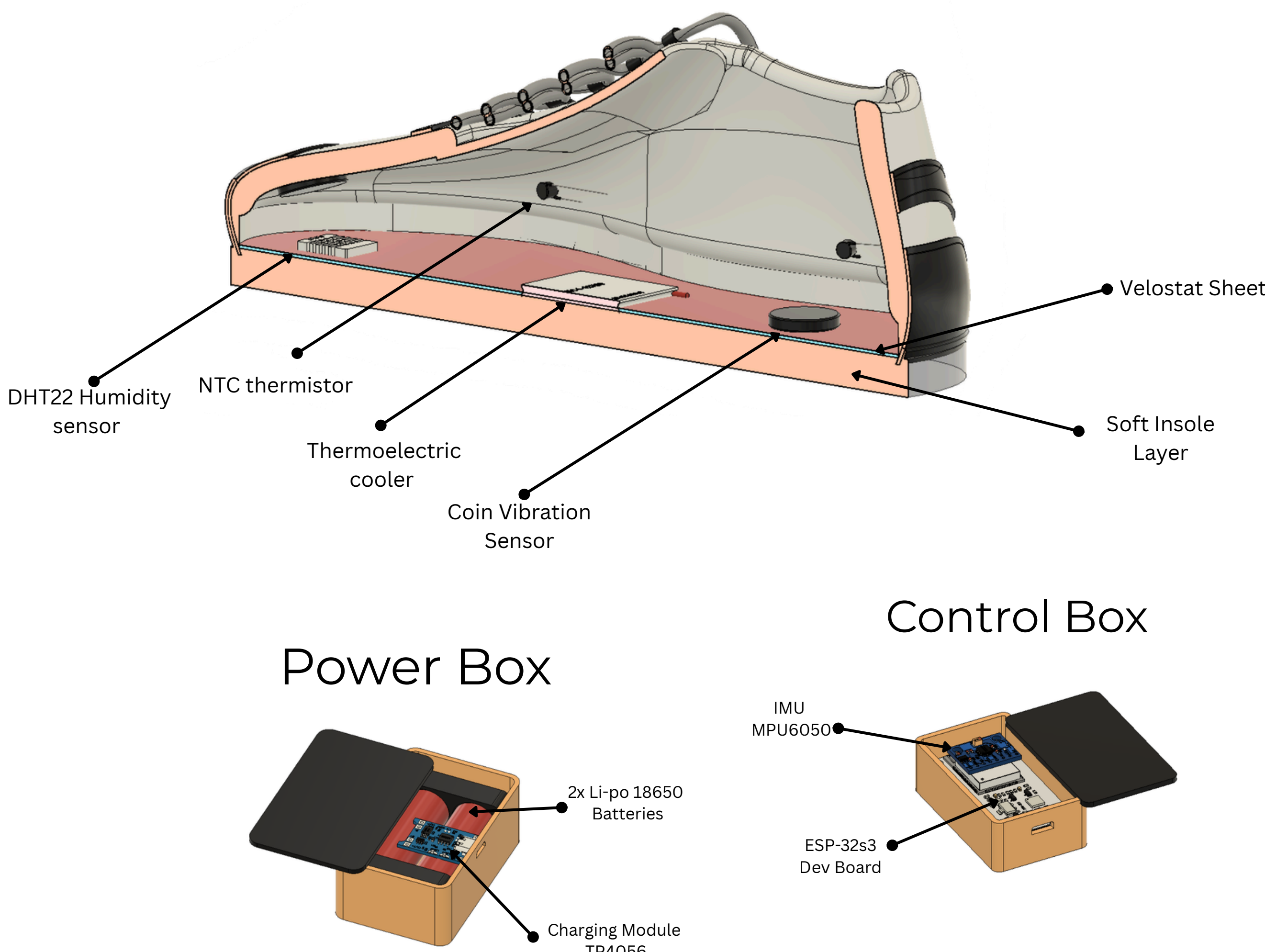
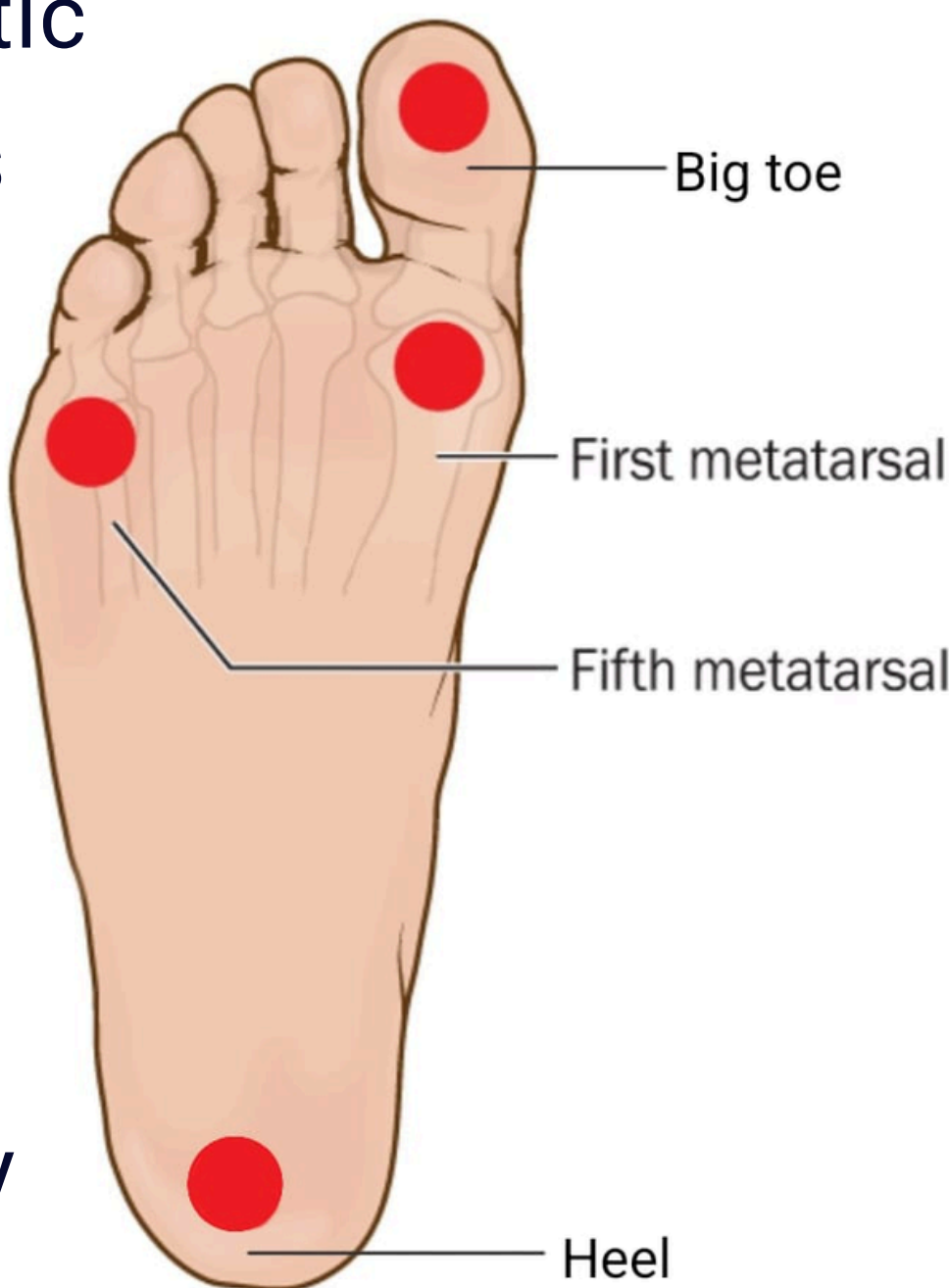


SMART DIABETIC SHOE : DETECTION AND PREVENTION SYSTEM

Part of : Measurements Systems and Sensors
Robotics , Autonomous and interactive systems program

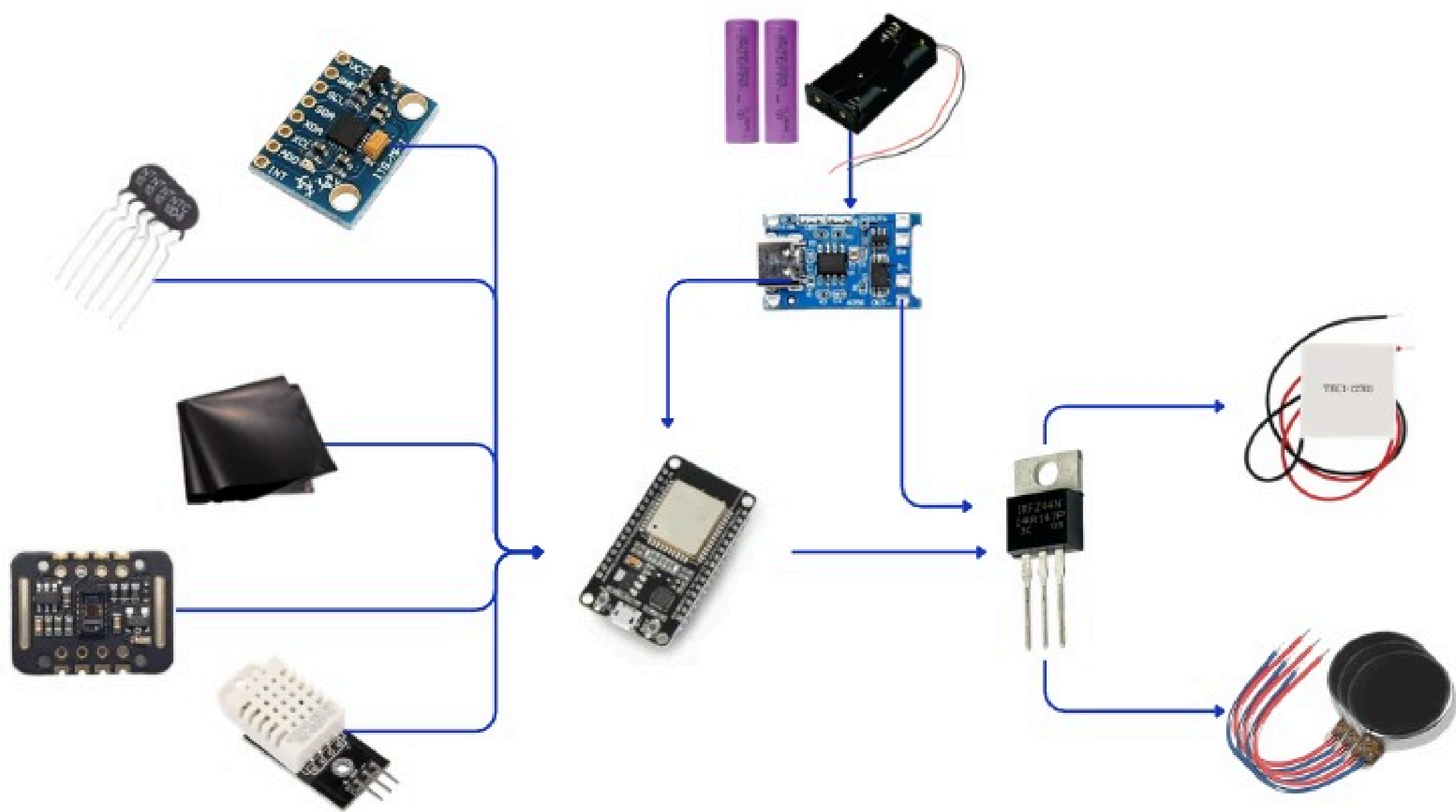
ABSTRACT

This project develops a smart wearable insole for early detection and prevention of diabetic foot ulcers. It monitors key factors such as pressure, temperature, humidity, gait, and blood flow in real time using multiple sensors. The system processes data through a microcontroller and provides immediate feedback via vibration alerts and cooling. The solution is low-cost, portable, and designed for continuous daily use, helping reduce ulcer risk through early intervention.



METHODOLOGY

The system integrates multiple sensors into a wearable insole connected to a microcontroller. Data is continuously collected and analyzed using a risk detection algorithm. When abnormal conditions are detected, the system activates vibration alerts and cooling. The prototype is tested and optimized for accuracy, response time, and usability.

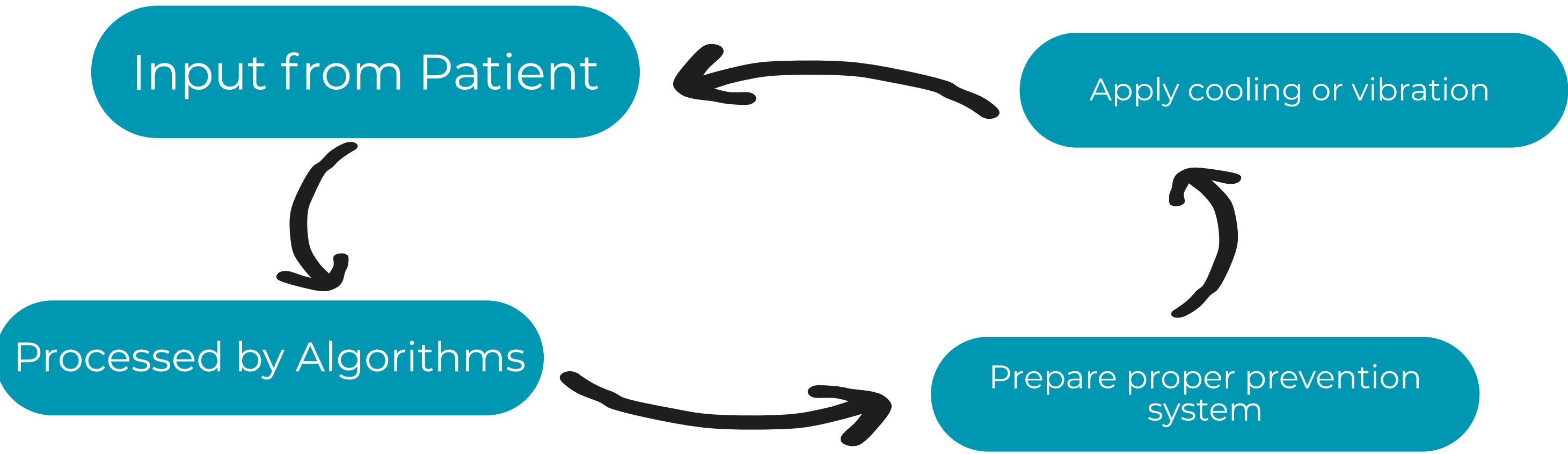
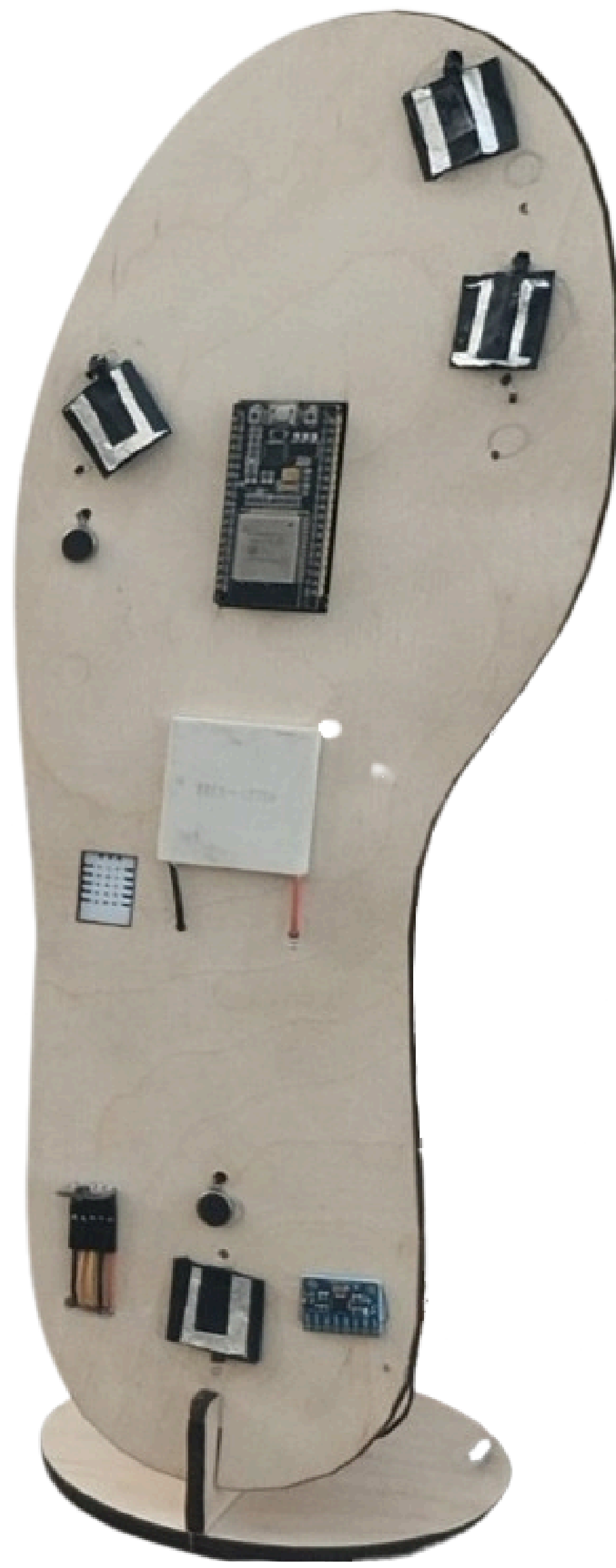


OBJECTIVES

- Monitor foot conditions in real time
- Calculate a risk index for ulcer formation
- Provide alerts and preventive actions
- Ensure a low-cost, practical design

Category	Details
Pressure & Temperature	
Top	Pressure: NONE (0) Temp: 23.5C
Middle Right	Pressure: Very High Temp: 23.5C
Middle Left	Pressure: NONE (0) Temp: 23.6C
Bottom	Pressure: NONE (0) Temp: 24.7C
Environment	
Humidity	48.0%
Heart Rate	
HR	71 bpm
Movement	
Walking	NO - stationary Magnitude: 1.006 g
Outputs	
Status	C1:off C2:on C3:off TEC:off Buzzer:off

PROTOTYPE



SUPERVISED BY



DR. SHAHINAZ HOSSAM



ENG. MOSTAFA RASHED

TEAM MEMBERS



HABIBA TAMER



YOUSEF ELAASER



MARYAM AHMED



MOHAMED OMAR